

Q-HAS: testing an Ising objective for adaptive mesh refinement in magnetohydrodynamics

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Abstract

Adaptive mesh refinement decides where a simulation spends resolution. Q-HAS (Quantum-Hierarchical Adaptive Steering) investigates whether a local quantum circuit can improve that decision in two-dimensional magnetohydrodynamics. Classical indicators initialise the qubit probabilities, while field gradients and circulation define an Ising Hamiltonian whose energy is minimised by QAOA. The central question is not only whether QAOA solves this Hamiltonian, but whether solving it gives a useful refinement decision. At 18 qubits, exhaustive enumeration separates these questions. On twelve snapshots from four scenarios, the exact decision has a lower mean F_1 than a classical threshold for both mapper implementations, before and after calibrating the threshold on other scenarios. In the simpler mapper, every one of the 24 instance–threshold pairs gives a uniform decision: refine everything or refine nothing. A larger comparison across four Reynolds numbers gives mean scores of 0.577 for the classical rule, 0.495 for a gradient boosted tree and 0.446 for QAOA; removing the coupling terms also improves the mean exact decision. The study identifies objective misalignment, readout dependence and task specification as distinct issues. It establishes neither a general limitation of quantum optimisation nor a performance advantage in a complete adaptive simulation. All circuits are simulated, and the results concern the tested mappings and refinement labels.

1 Introduction

Magnetohydrodynamics (MHD) describes the coupling between fluid motion and magnetic fields in conducting fluids and plasmas [1]. Shear layers roll up into vortices, current sheets develop fine structure, and magnetic reconnection changes field connectivity. These structures are local, but failing to resolve them can affect the evolution of the whole system. Uniform resolution is expensive precisely because the smallest relevant scales need not occupy the whole domain.

Adaptive mesh refinement (AMR) increases resolution only where it is needed [2, 3]. The difficulty is deciding where. A threshold on an instantaneous gradient or error indicator is inexpensive, but does not directly optimise the accuracy and cost of the future trajectory. This motivates criteria that use spatial context or temporal information. Classical learning approaches already investigate this possibility, including reinforcement learning for refinement policies and anticipatory mesh adaptation [4, 5]. Anticipation is therefore an objective for Q-HAS, not a capability unique to a quantum circuit.

The starting intuition of Q-HAS is simple: MHD structures have local interactions, so an Ising Hamiltonian on a coarse grid might express which neighbouring regions should be refined together. The classical simulation still integrates the equations. The quantum component only chooses a refinement map. QAOA alternates evolution under this cost Hamiltonian with a mixer and adjusts the circuit parameters through classical optimisation [6]. This follows the general

use of Ising objectives for discrete optimisation [7], with coefficients constructed here from the plasma fields.

There are two separate requirements. The optimiser must find a low energy configuration, and low energy configurations must correspond to good refinement decisions. The first does not imply the second. An accurate solver of a poorly aligned objective can make the decision worse, while an approximate solver can retain a useful part of its classical initialisation. This distinction is the main subject of the paper.

The problem sizes make that distinction testable. A periodic grid of $d \times d$ blocks carries $2d^2$ qubits: 8 qubits at $d = 2$ and 18 at $d = 3$. Their 256 and 262 144 configurations can be enumerated classically. The study consequently tests a decision rule, not a complexity advantage. Larger grids are considered only through classical optimisation of the same energy.

The contribution is an explicit mapping, an evaluation against exact and classical references, and a diagnosis of where the mapping fails. The current twelve-snapshot experiment also separates two possible confounders: the threshold around which the bias is centred and the rule used to turn circuit marginals into decisions. A broader comparison tests transfer between four scenarios. The conclusion is negative for the tested objective, but deliberately narrower than a claim that quantum or learned AMR cannot work.

2 From the plasma fields to a refinement decision

2.1 The physical fields and the coarse graph

The implementation evolves incompressible, viscous and resistive MHD on the periodic domain $\Omega = [0, 2\pi]^2$. With constant density and the magnetic field expressed in velocity units, the continuum equations are

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \Pi + (\mathbf{B} \cdot \nabla) \mathbf{B} + \nu \nabla^2 \mathbf{v}, \quad \nabla \cdot \mathbf{v} = 0, \quad (2.1)$$

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \quad \nabla \cdot \mathbf{B} = 0. \quad (2.2)$$

Here Π includes magnetic pressure. The numerical parameters use $\nu = Re^{-1}$ and $\eta = Rm^{-1}$. The reference trajectories considered below have $Re = Rm$.

The domain is divided into d^2 blocks. Each block has one horizontal and one vertical edge qubit, with periodic identification at the boundary. This gives a graph on which neighbouring edge qubits can be coupled and four edge qubits can form a plaquette. Field values used by the decision study are averaged within each block; the classical score is instead reduced by its maximum, so a narrow feature is not lost by averaging its indicator.

2.2 The classical score and its encoding

Four indicators are computed on the fine grid: the magnitude of vorticity, velocity divergence, current density, and a second-difference indicator on $|\mathbf{B}|$ inspired by Löhner [8]. Each is normalised by its spatial maximum when that maximum is resolved. Their combination is

$$s(x, y) = \left(\frac{s_\omega^2 + s_{\text{div}}^2 + s_J^2 + s_{\text{Löh}}^2}{4} \right)^{1/2}, \quad s_i = \max_{(x, y) \in b_i} s(x, y). \quad (2.3)$$

This is the particular classical baseline used here, not a claim that production MHD codes all use this combination. The root mean square reduces the weight of a single indicator relative to taking a maximum; it does not require all indicators to agree. In an incompressible calculation, velocity divergence is principally a numerical diagnostic rather than an independent physical instability.

Both edge qubits attached to block i start with the probability s_i of measuring 1. For edge e , attached to block $i(e)$,

$$|\psi_0\rangle = \bigotimes_e \left(\sqrt{1 - s_{i(e)}}|0\rangle + e^{i\phi_e} \sqrt{s_{i(e)}}|1\rangle \right), \quad \theta_e = 2 \arcsin \sqrt{s_{i(e)}}. \quad (2.4)$$

The phase records the change in a stress indicator Φ_e between two field snapshots:

$$\phi_e = \frac{\pi}{2} \tanh \left(b_\phi \frac{\Phi_e(t) - \Phi_e(t^-)}{\langle |\Delta \Phi| \rangle + \epsilon} \right). \quad (2.5)$$

The stress indicator combines normal and tangential field jumps and current variations. It is a feature of the fields, not the conservative flux of the MHD integrator. The study uses the preceding saved snapshot for t^- ; the adaptive pipeline also supports a running reference. These are different temporal sampling choices.

Equation (2.4) retains classical information in the initial state. It is related in motivation to initialising quantum optimisation from a classical solution [9], but uses the standard X mixer, not the specialised mixer of that work. Neither improved convergence nor protection against vanishing gradients follows from the encoding alone.

2.3 The Ising objective

The cost operator has the form

$$H = \sum_e h_e Z_e + \sum_{(e,f) \in \mathcal{E}} C_{ef} Z_e Z_f + \sum_{p \in \mathcal{P}} (K_p + K_{X,p}) \prod_{e \in p} Z_e. \quad (2.6)$$

The term K_X is optional in the architecture and enabled in the principal experiments. The bias is

$$h_e = \alpha_z (s_{i(e)} - \tau), \quad \alpha_z \geq 0. \quad (2.7)$$

Since $Z|1\rangle = -|1\rangle$, a score above τ favours refinement. The gradient coefficients satisfy $C_{ef} \leq 0$ and favour equal neighbouring spins. The plaquette coefficients are also nonpositive, but favour *even parity*, not equality of all four spins. These signs are part of the definition of the objective.

Two coefficient mappings are tested. $V1$ is the more structured mapping used by the adaptive pipeline. Velocity and magnetic jumps are weighted by local Reynolds factors and strain gates. A Gaussian window concentrates the pair couplings near the threshold. Rotation and current gates weight the circulation terms. Its coefficients have adjustable sensitivities, evaluated here at the repository's reference values rather than after a new training campaign. Appendix A gives the implemented structure and settings.

$V2$ removes these adjustable gates. Let j_{ef} be the norm of the velocity and magnetic jumps across a coarse interface, and let $\hat{\omega}_p$, $\hat{J}_p$ and $\hat{X}_p$ denote separately normalised magnitudes of discrete curl, current and $\max(0, -\det \nabla \mathbf{B})$. The current $V2$ mapping is

$$C_{ef} = -2 \frac{j_{ef}}{\max j}, \quad K_p = -\frac{\hat{\omega}_p + \hat{J}_p}{M}, \quad K_{X,p} = -\frac{\hat{X}_p}{M}, \quad (2.8)$$

$$M = \max_p (\hat{\omega}_p + \hat{J}_p + \hat{X}_p), \quad \alpha_z = 0.1 \max\{|C|, |K + K_X|\}. \quad (2.9)$$

An absent or unresolved signal contributes zero; a zero denominator is replaced by 1. The precise roundoff guards are in the source. With these conventions, $|C| \leq 2$ and $|K + K_X| \leq 1$. $V2$ has fixed design weights, but it still has a threshold and variational circuit parameters. Calling the whole procedure parameter free would therefore be misleading.

These terms express preferences about refinement maps. They are not the physical energy of the plasma and do not constitute an error estimator derived from Eqs. (2.1)–(2.2). A large

circulation coefficient can indicate a relevant region without its preferred spin parity giving the correct refinement decision. This is the assumption that needs testing.

The coefficients are not independent classifiers of instability type either. In V1 the strain and rotation gates are complementary functions of one scalar; in V2, curl and current magnitudes are added into one plaquette coefficient. The determinant signal adds information about hyperbolic magnetic geometry, but a negative determinant away from a magnetic null does not by itself establish reconnection.

2.4 The circuit and the two readout rules

For p layers, the state and variational energy are

$$|\psi(\boldsymbol{\beta}, \boldsymbol{\gamma})\rangle = U_p \cdots U_1 |\psi_0\rangle, \quad U_k = e^{-i\beta_k \sum_e X_e} e^{-i\gamma_k H}, \quad \mathcal{E} = \langle \psi | H | \psi \rangle. \quad (2.10)$$

The mixer generator is $\sum_e X_e$; its unitary factors into single-qubit rotations. The implementation minimises $\mathcal{E}$ using COBYLA [10] and constrains $|\beta_k| \leq \pi/(4p)$. The bound is a design choice, not a QAOA theorem.

Let $m_{i,h}$ and $m_{i,v}$ be the measured probabilities of 1 for the two edge qubits at block i . The original study and the adaptive pipeline use different decisions:

$$\hat{y}_i^{\text{maj}} = \mathbf{1}\{m_{i,h} > 1/2 \text{ or } m_{i,v} > 1/2\}, \quad (2.11)$$

$$\hat{y}_i^{\text{dep}} = \mathbf{1}\{(m_{i,h} + m_{i,v})/2 \geq \tau\}. \quad (2.12)$$

The first is called the majority readout because each edge is thresholded at $1/2$; the block then takes their logical OR. The second is the deployed readout. They are not equivalent, even at $\tau = 1/2$ for general unequal marginals. Before any QAOA layer, $m_{i,h} = m_{i,v} = s_i$, so Eq. (2.11) gives the score threshold $1/2$, whereas Eq. (2.12) gives the deployed threshold. The classical baseline uses $s_i > \tau$; the different convention at equality is retained explicitly.

For a classical bit configuration, the study defines the exact decision by the OR of its two edge bits, consistently with Eq. (2.11). The experiment with Eq. (2.12) changes the QAOA readout and duplicates each resulting block decision onto its two edge bits when assigning it an energy. The stored exact reference retains its original OR convention. It is therefore a readout sensitivity test, not a complete reevaluation of every solver through the deployed decoder. An additional enumeration check in Table 1 applies Eq. (2.12) to the exact states themselves. This distinction matters when $\tau > 1/2$ and the two bits disagree.

The adaptive architecture recursively applies Eq. (2.12) to smaller patches, with neighbouring information supplied by a halo and rules for patch persistence. The main results below isolate the periodic decision at one level. They do not measure the performance of that full recursion.

2.5 What is known directly from the circuit

All terms of H are diagonal. If the mixer angles vanish, then

$$|\psi(\mathbf{0}, \boldsymbol{\gamma})\rangle = e^{-i(\sum_k \gamma_k)H} |\psi_0\rangle, \quad |\langle b | \psi(\mathbf{0}, \boldsymbol{\gamma}) \rangle|^2 = |\langle b | \psi_0 \rangle|^2. \quad (2.13)$$

Thus changing the cost angles alone changes neither the measured probabilities nor $\mathcal{E}$. The cost Hamiltonian influences the decision through its interaction with the mixer. It does not follow that the Hamiltonian is irrelevant, or that turning on the mixer must improve the decision. The phase in Eq. (2.5) likewise supplies temporal information to the initial state, but not to the energy in Eq. (2.6). An exact ground state is independent of that phase.

3 Evaluation protocol

3.1 Reference fields and the task being scored

Four scenarios are used: a Harris current sheet, a Kelvin–Helmholtz shear layer, an MHD rotor and an Orszag–Tang vortex [11]. These are the incompressible implementations provided by the repository, not interchangeable with every compressible benchmark bearing the same name. The principal decision experiments use 96×96 fine grids and $d = 3$ coarse blocks per direction.

The reference solver uses fourth-order centred spatial differences, RK4 and a velocity projection after each step [12]. Its curl form for magnetic induction preserves the corresponding discrete magnetic divergence without a separate magnetic projection. Importantly, tests of the *complete* time step observe approximately first-order temporal convergence with the final velocity projection. RK4 alone does not justify a fourth-order accuracy claim for the full algorithm. The saved fine-grid trajectories are numerical references, not exact continuum solutions.

For $U = (v_x, v_y, B_x, B_y)$, the static score of a block is the information lost when its fields are replaced by their block means:

$$e_i(t) = \frac{\left[|b_i|^{-1} \sum_{x \in b_i} \|U(x, t) - \bar{U}_i(t)\|_2^2\right]^{1/2}}{\left[|\Omega_h|^{-1} \sum_{x \in \Omega_h} \|U(x, t)\|_2^2\right]^{1/2}}, \quad y_i(t) = \mathbf{1}\{e_i(t) \geq q_{0.75}\}. \quad (3.1)$$

Here Ω_h is the fine grid and $q_{0.75}$ is the 75th percentile over all blocks and saved times of that trajectory. The denominator is set to 1 for an effectively zero field. Ties can make the positive fraction exceed 25%. This label measures instantaneous spatial variation. It is not the reduction in future simulation error obtained by refining the block. Its trajectory percentile also uses future reference data to define the offline target; it is not an online refinement threshold.

Decision quality is measured by

$$F_1 = \frac{2TP}{2TP + FP + FN}, \quad (3.2)$$

with value zero when the denominator vanishes. The small solver panel averages F_1 over snapshots. The larger scenario comparison pools the blocks of the held-out snapshots before computing F_1 . These are different summaries and are not compared as if they had identical weighting.

3.2 Exact optimisation and threshold recalibration

The main diagnostic contains twelve snapshots: three from each scenario at $Re = Rm = 400$. Two mappers, two threshold choices and two QAOA readouts give eight experiment settings. The original thresholds are 0.1496 for V1 and 0.15 for V2. The alternative threshold maximises the training F_1 over a specified finite grid of candidate thresholds, using only the three other scenarios. Calibration uses ten snapshots per scenario and Reynolds number, at $Re \in \{400, 800, 1200, 1600\}$. The held-out scenario contributes no scores or labels to this fit. The resulting thresholds are 0.5208, 0.6000, 0.5923 and 0.5100, respectively, for Harris, Kelvin–Helmholtz, rotor and Orszag–Tang.

Each setting compares nine solver configurations on the same Hamiltonians: exhaustive enumeration, simulated annealing from random and classical starts, greedy descent, the unoptimised classical decision, QAOA at $p = 1, 2, 3$, and an additional Aer configuration at $p = 3$. Annealing uses five restarts and 500 sweeps. QAOA uses a budget of 60 optimisation evaluations, 4096 measurement shots and seed 0. The principal backend is named `state_vector`; the additional backend is named `aer`. Both still sample the readout. Their names should not be interpreted as an exact-marginal versus sampled-marginal experiment. The circuits run through Qiskit and Aer [13, 14], without a hardware noise model.

For decoded spins $z_e \in \{-1, 1\}$, the energy and normalised gap are

$$E(z) = \sum_e h_e z_e + \sum_{(e,f)} C_{ef} z_e z_f + \sum_p (K_p + K_{X,p}) \prod_{e \in p} z_e, \quad g(z) = \frac{E(z) - E_*}{\max(|E_*|, 10^{-12})}. \quad (3.3)$$

Enumeration supplies E_* . A decoded optimum hit means $E(z) \leq E_* + 10^{-6} \max(|E_*|, 1)$. This is the energy of a thresholded configuration, not the variational energy $\mathcal{E}$, the probability of sampling a ground state, or the fidelity of the quantum state with a ground state. The distinction applies to the majority readout as well as the deployed readout.

For each solver, g and F_1 are averaged over the twelve instances. Their Spearman correlation is then computed over the nine solver configurations. This is a descriptive comparison of solvers on shared data. The nine entries are neither nine independent physical experiments nor nine independent implementations; in particular, several give tied results.

3.3 Comparison across scenarios and Reynolds numbers

The larger experiment uses four scenarios, four Reynolds numbers and ten snapshots per pair, giving 160 snapshots in total. Each fold holds out one scenario at every Reynolds number and fits its classical controls on the other three. There are 40 held-out snapshots, or 360 block decisions, per fold. Reynolds number is varied within training and evaluation; it is not an unseen extrapolation axis.

The first control thresholds Eq. (2.3), with its threshold fitted on the training pool. The second is a gradient boosted tree classifier [15, 16]. Its nine features per block are the score, $|\mathbf{v}|^2$, $|\mathbf{B}|^2$, the two curl magnitudes, the squared velocity and magnetic gradient magnitudes, the magnetic Jacobian determinant, and Re . It uses at most 300 boosting iterations, learning rate 0.05, at most 31 leaves, at least 20 samples per leaf, and early stopping with a 10% internal validation subset. Its decision threshold is also chosen on training predictions. That internal split is not a separate trajectory validation study.

The tree is a flexible classical control, not an upper bound on all learnable functions. The repository also supports adding neighbouring feature vectors, but this particular comparison uses the nine local features; a claim about a tested neighbourhood-augmented tree would be inaccurate.

QAOA uses V2, $p = 2$, 60 optimisation evaluations, 4096 shots and the majority readout. Its Hamiltonian threshold remains 0.15. Exact and QAOA decisions are also obtained after removing all ZZ and $ZZZZ$ terms while keeping the already computed Z bias and initial state. The main comparison therefore includes separately calibrated classical controls but a fixed-threshold Hamiltonian. Section 4.1 addresses this mismatch through the smaller recalibration experiment.

The stored 95% intervals use 1000 paired bootstrap resamples of complete held-out snapshots, retaining all nine blocks in each sampled snapshot. This respects within-snapshot spatial dependence, but not serial dependence between snapshots of a trajectory. The intervals are conditional summaries of this dataset, not confidence guarantees over independent physical initialisations. They do not include training uncertainty or repeated circuit seeds. No multiplicity-adjusted confirmatory claim is made.

4 Results: solving the objective and solving the task

4.1 Recentring the threshold does not repair the exact decision

The original threshold was inherited from a different objective, balancing simulation error and refinement cost. It is much lower than the threshold fitted to the static label in Eq. (3.1). It was therefore necessary to test whether the apparent failure of the Hamiltonian was only a threshold mismatch.

Mapper	Threshold	Classical	Exact OR	Exact dep.	All	None
V1	Original	0.468	0.437	0.437	9/12	0/12
V1	Calibrated	0.479	0.405	0.298	1/12	2/12
V2	Original	0.468	0.386	0.386	12/12	0/12
V2	Calibrated	0.479	0.108	0.108	4/12	8/12

Table 1: Mean snapshot F_1 and exact decisions in the current twelve-instance panel. Calibration excludes the evaluated scenario. All 48 instance–Hamiltonian pairs have a unique enumerated optimum. “All” and “None” count uniform masks under the stored OR decoder. “Exact dep.” is an additional enumeration check with Eq. (2.12). Sources and the verification script are specified in Appendix B.

Table 1 gives the direct comparison. For both mappers and both threshold choices, the mean exact F_1 is below that of the classical rule using the same threshold. These are four distinct Hamiltonian settings, not eight independent replications: changing the QAOA readout does not change the exact reference.

The qualification “in mean” matters. Grouped by trajectory, the stored OR reference wins one of the sixteen comparisons, ties four and loses eleven. The exception is calibrated V1 on Orszag–Tang, with $F_1 = 0.450$ against 0.250 for the classical rule. It would be incorrect to turn the aggregate result into “the exact decision never wins on any scenario.” Applying the deployed decoder to the enumerated states changes nine calibrated V1 masks and lowers their mean F_1 to 0.298; the other three settings are unchanged. The central result therefore also holds with that common deployed decoder. V1 recalibration moves its Gaussian coupling window and the resulting bias scale, so it is not an intervention on the bias alone. V2 gives the cleaner threshold-only test.

4.2 Why a spatial objective can return a uniform mask

In V2, the exact mask is uniform in all 24 instance–threshold pairs. At the original threshold, it always refines everything. After recalibration it refines nothing in eight instances and everything in four. A unique optimum can therefore be a uselessly uniform decision; uniqueness and spatial discrimination are different properties.

The mechanism is visible directly in Eq. (2.6). The all-zero and all-one bit configurations minimise every nonpositive pair and plaquette term simultaneously. Their interaction energies are equal, so

$$E(+1) - E(-1) = 2 \sum_e h_e = 2\alpha_z \sum_e (s_{i(e)} - \tau). \quad (4.1)$$

Among those two states, the sum of the biases decides which one wins. If leaving this uniform pair costs more interaction energy than it saves in bias energy, local score differences cannot change the ground state. The experiment finds precisely a uniform optimum for V2 on this panel. Equation (4.1) explains the competition; it is not a proof that every instance of the model must be uniform.

This also clarifies the role of the control without couplings. Provided the biases are nonzero and retained, independent minimisation gives

$$\arg \min_{z_e \in \{-1, 1\}} h_e z_e = \begin{cases} -1, & s_{i(e)} > \tau, \\ +1, & s_{i(e)} < \tau. \end{cases} \quad (4.2)$$

Thus the bias alone recovers the classical threshold away from ties. Any improvement from the couplings must come from correcting this rule in the right places, not merely from changing it. Coefficient pruning and zero biases can create exceptions to the identity and must be kept in the numerical comparison.

Mapper	Threshold	QAOA readout	QAOA F_1	Decoded hits	$\rho(g, F_1)$
V1	Original	Majority	0.503–0.519	0/12	+0.891
V1	Original	Deployed	0.468–0.488	6/12	+0.632
V1	Calibrated	Majority	0.515–0.519	0/12	+0.863
V1	Calibrated	Deployed	0.479–0.507	0/12	+0.863
V2	Original	Majority	0.334–0.411	0–2/12	–0.148
V2	Original	Deployed	0.356–0.362	7–8/12	–0.130
V2	Calibrated	Majority	0.341–0.357	2–3/12	+0.979
V2	Calibrated	Deployed	0.265–0.294	3–5/12	+0.828

Table 2: Current solver panel. F_1 and decoded-hit ranges span QAOA depths $p = 1, 2, 3$, not uncertainty intervals. Each correlation uses the nine solver averages, including the additional Aer configuration. The hit test concerns Eq. (3.3), not ground-state sampling probability. Both backend configurations sample their readout.

4.3 QAOA performance depends on what is decoded

Table 2 reports both readouts. With the majority rule, no V1 QAOA configuration reaches a decoded optimum on any of the twelve snapshots. With the deployed rule and original threshold, six of twelve decoded configurations do. This change does not show that the underlying variational optimisation improved: the mapping from marginals to a classical configuration changed.

The initial score already explains much of V1’s result. In the original majority panel, its QAOA F_1 equals that of the unoptimised score threshold $s_i > 0.5$ on eleven of twelve instances at each depth. This is an equality of scores, not a verified equality of masks. The majority rule itself changes 34 of the 108 block decisions relative to the original classical threshold before any optimisation. A comparison that ignores this change can credit the circuit for a different decoder.

Recentring gives positive energy-gap versus F_1 correlations for both mappers and both readouts. Lower decoded energy is not associated with a better task score in these settings. However, the current V2 mapping at the original threshold has a slightly negative correlation. The historical +0.870 value used an earlier normalisation and is not a current result. The robust observation is the exact-versus-classical comparison in Table 1, not a universal sign for the correlation.

Some approximate V1 decisions do exceed the classical mean on this small panel. That is reported rather than removed: calibrated V1 with the deployed readout reaches 0.507, against 0.479 for the classical rule. It does not establish a reproducible gain across independent trajectories or seeds, nor does it repair the alignment of the exact objective. Approximate decision quality and exact objective quality remain separate questions.

4.4 The larger comparison favours the classical threshold

Table 3 gives the larger V2 experiment. The classical threshold beats or ties QAOA and the fitted tree in every held-out scenario. The arithmetic means across the four folds are 0.577, 0.495 and 0.446 for the classical rule, tree and QAOA, respectively. This is a result about these three configurations, not a proof that a threshold is the best possible model.

The stored paired intervals for QAOA minus classical are $[0, 0]$ for Harris, $[-0.154, -0.143]$ for Kelvin–Helmholtz, $[-0.337, -0.119]$ for the rotor and $[-0.246, -0.079]$ for Orszag–Tang. Their interpretation is subject to the dependence limitations in Section 3.3. A zero-width interval on F_1 does not prove that two masks are identical.

On the exact reference, adding the couplings wins no fold: Harris and rotor lose 0.167 and 0.218, while the other two tie in F_1 . Their conditional bootstrap intervals for full minus Z are $[-0.167, -0.167]$, $[0, 0]$, $[-0.263, -0.169]$ and $[0, 0]$. The QAOA ablation is less uniform:

Held-out scenario	Classical	Tree	QAOA	Exact	QAOA Z	Exact Z
Harris	0.667	0.667	0.667	0.500	0.667	0.667
Kelvin–Helmholtz	0.571	0.571	0.423	0.421	0.422	0.421
Rotor	0.690	0.456	0.476	0.351	0.485	0.569
Orszag–Tang	0.379	0.286	0.220	0.349	0.374	0.349
Mean	0.577	0.495	0.446	0.405	0.487	0.501

Table 3: Pooled held-out F_1 , 40 snapshots per fold, $Re = Rm \in \{400, 800, 1200, 1600\}$, $N = 96$, $d = 3$. The Z columns remove couplings while preserving the computed bias. Classical and tree thresholds are fitted on training scenarios; V2 retains threshold 0.15 and QAOA uses Eq. (2.11). This is a separate experiment from Tables 1–2.

its rotor difference is unresolved by the stored interval, and removing the couplings improves Orszag–Tang. It would be too strong to say that every coupling effect is harmful in every circuit.

QAOA exceeds the exact reference with stored intervals strictly above zero on Harris and rotor, while Orszag–Tang favours the exact reference. Kelvin–Helmholtz remains unresolved: its interval starts at zero. The recorded fraction of bootstrap differences above zero is a bootstrap frequency, not a conventional hypothesis-test p -value.

4.5 What the size scan adds

Earlier V1 ablations give mean F_1 values of 0.333 versus 0.333 at 8 qubits, 0.520 versus 0.552 at 32, and 0.592 versus 0.648 at 128, for the full objective versus its bias alone. They support investigating the coupling penalty beyond the smallest graph, but have a different evidential status. Only the 8-qubit row is exhaustive; the two larger rows use greedy descent. The 8-qubit control agrees with exhaustive decisions in 75% of the recorded cases, not universally. Moreover, those artifacts predate later V1 changes and record uncommitted source changes at their creation. They are historical diagnostics, not certified current ground-state bounds at 128 qubits. The present conclusion does not depend on them.

5 Task specification and numerical checks

The static task is closely related to the features from which the classical rule is constructed. At $N = 96$ and $d = 16$, the repository reports areas under the ROC curve of 1.000, 0.997, 0.948 and 0.592 for the classical score on Harris, Kelvin–Helmholtz, rotor and Orszag–Tang. These are ranking diagnostics at a different block size, not the F_1 scores of Table 3. They show that little ranking information is missing on three scenarios, but do not guarantee that one transferable threshold attains a perfect decision.

A more predictive target measures what happens *after* information is removed. Replace only block i by its mean, evolve this modified field and the unmodified field with the same saved time-step sequence, and compare their whole-domain fields after a horizon Δt . Denoting the numerical evolution by $\mathcal{S}_{\Delta t}$ and this block replacement by $\mathcal{C}_i$, the dynamic error is

$$d_i(\Delta t) = \frac{\|\mathcal{S}_{\Delta t}(\mathcal{C}_i U) - \mathcal{S}_{\Delta t}(U)\|_{\text{RMS}}}{\|U\|_{\text{RMS}}}, \quad t_x = \frac{2\pi/d}{v_{\text{RMS}} + B_{\text{RMS}}}. \quad (5.1)$$

The comparison with the static error concerns rankings, so the constant conversion between block and whole-domain normalisations at the initial time is immaterial. Freezing the time-step sequence avoids counting different adaptive time stepping as sensitivity to the removed information.

At $d = 8$, $N = 96$ and $Re = 400$, the horizon 0.1 gives almost the same rankings as the static error. One estimated patch crossing time is roughly four to nine times longer. At that horizon,

five snapshots per scenario still give correlations at least 0.965 for Harris and Kelvin–Helmholtz. The earliest rotor and Orszag–Tang snapshots instead fall to 0.817 and 0.658. The horizon can expose information absent from the static ranking, but not uniformly across the panel. Even when rankings remain similar, spatially varying amplification can carry different information. No result here shows that the encoded phase successfully exploits it.

Checking the implementation was also essential. The audit log documents more than 190 corrected contract defects; the relevant lesson is not the count but what was compared. A derivative must use the declared axis convention. Divergence must be checked with the discrete operator that the evolution preserves. A contracted halo must correspond to the qubit and coefficient it replaces. A coefficient that is nonzero on a constructed field must also remain active at the resolution actually used by the simulation.

Those checks changed the interpretation of earlier results. In particular, a defect in the selection of patch data affected Q-HAS below the root of the adaptive tree. Historical results from the full adaptive simulation cannot be retained as a controlled performance comparison after that correction. They are not used as evidence of either an advantage or a disadvantage here. Likewise, correcting numerical contracts does not prove that all remaining modelling choices are physically adequate.

6 Discussion

The initial idea was that local spatial structure could improve a refinement decision beyond a cheap score. The tested Hamiltonian does encode spatial structure. The difficulty is what minimising that encoding asks the decision to do. Negative pair couplings encourage agreement, and negative plaquette couplings encourage parity. Neither preference is automatically aligned with the blocks whose unresolved variation is largest.

The exact reference makes this failure visible without blaming the variational optimiser. On the current small panel, the mapped ground-state decision is worse in mean than thresholding the same score, even after recalibration. For V2, the ground state is always uniform. Improving the optimiser alone cannot improve the decision obtained from these particular ground states. This is a statement about the tested objective and decoder, not a proof that no different coupling strength or different objective could work.

Conversely, an approximate QAOA decision can exceed the score of the exact reference without demonstrating an advantage of solving the Ising objective. Its initial state carries the classical score and temporal phase, and its decoder may apply a more favourable effective threshold. A useful comparison must therefore separate initialisation, readout, mixer and cost terms. Equation (2.13) motivates a mixer-only control, but the main solver panel does not contain a comparably selected mixer-only rule. Optimising a zero Hamiltonian would provide no objective for choosing that rule; an explicit angle-selection protocol would be needed.

The tree result has a similarly limited interpretation. A more flexible model fails to beat the tested threshold in the larger comparison. This does not rule out learning: it concerns one feature set, model and calibration procedure, on four scenarios. Published classical work on adaptive and anticipatory refinement addresses different tasks [4, 5]. The result here instead suggests that reproducing an instantaneous variation label may be a poor way to demonstrate predictive value.

Several limits remain. The twelve-instance panel has four trajectories, not twelve independent physical realisations. The larger panel does not extrapolate to unseen Reynolds numbers, and its snapshot bootstrap does not account for temporal dependence. QAOA uses fixed budgets and one recorded seed; the experiments do not exhaust its optimisation strategies. V1 has not undergone the complete new hyperparameter campaign envisaged by the repository. The larger V2 comparison has not been repeated with the recalibrated Hamiltonian threshold and a common deployed decoder for every arm. Finally, the numerical reference has a documented

temporal-order limitation, and no current complete adaptive campaign establishes a reduction in simulation error at matched computational cost.

These are specific missing experiments rather than reasons to discard the exact diagnostic. The most useful next step is to define the desired refinement benefit, align the objective with it, and test the exact decision before paying for a more expensive optimiser. For a predictive task, Eq. (5.1) supplies one possible target, with a physically motivated horizon. A complete comparison would then keep the decoder and computational budget explicit and evaluate independent trajectories. Nothing in the present results establishes that this redesigned task requires a quantum solver.

7 Conclusion

Q-HAS provides a concrete way to encode MHD indicators into a small quantum circuit and an Ising objective for refinement. The evaluation shows why constructing such an objective and minimising it are not enough. At an exactly enumerable size, the ground-state decision is worse in mean than a classical threshold in every tested mapper–threshold setting. Recentring the threshold does not repair this result; in V2 it changes which uniform mask is selected. In the larger comparison, the classical rule also beats or ties QAOA and a gradient boosted tree in all four held-out scenarios.

The result is a diagnosis of the tested mapping, not a general rejection of quantum optimisation or learned refinement. The contribution is the separation of objective quality, optimisation quality and readout, supported by exact references and traceable numerical artifacts. For this architecture, the priority is to make the objective express the refinement benefit actually sought. A better optimiser cannot substitute for that step.

Code and data availability

The source and numerical artifacts are available in [armandld/BA_Proj](#). This manuscript identifies the inspected repository revision as [f575f1d87023](#). Experiment revisions and the distinction between recomputing summaries and rerunning circuits are given in Appendix B. No new QAOA training campaign is implied by the preparation of this manuscript.

Use of AI assistance

This report was written with the assistance of artificial intelligence.

A Details of the structured mapper

This appendix records the current V1 construction rather than importing the older report’s simplified formulas. The source [HamiltParams.py](#) is the reference for finite-difference stencils and numerical guards. Its scalar gates are clipped in the implementation:

$$f(r; \gamma) = \min \left(10, \begin{cases} r, & r \leq 1, \\ 1 + \gamma \log r, & r > 1, \end{cases} \right), \quad (\text{A.1})$$

$$T(a; a_c, b) = \min\{10, b \max(0, a/(a_c + \epsilon) - 1)\}. \quad (\text{A.2})$$

The ratio r uses a local cell Reynolds or magnetic Reynolds number, not merely the global input Re . These scale factors are modelling choices, not derived critical thresholds for each instability.

The repository uses the sign convention $Q = (\omega^2 - \text{strain}^2)/2$, so positive Q denotes rotation dominance. With that convention, the topology gates are

$$g_{\text{strain}}(Q) = \frac{1}{1 + e^{\kappa Q/Q_c}}, \quad g_{\text{rot}}(Q) = \frac{1}{1 + e^{-\kappa Q/Q_c}}, \quad g_{\text{mag}}(J) = \frac{1}{1 + e^{-\kappa(|J|/J_c - 1)}}. \quad (\text{A.3})$$

This is an Okubo–Weiss-type diagnostic [17] with an explicitly stated sign convention. The first two gates sum to 1. That complementarity does not remove the distinct velocity, magnetic and curvature inputs multiplying them.

For an interface, let Δv , ΔB be vector-jump magnitudes and $\Delta_2 v$, $\Delta_2 B$ the implementation's second-derivative contrasts. Let T_v, T_B, T_{2v}, T_{2B} stand for Eq. (A.2) on those signals, with their respective thresholds. Then

$$C = -g_{\text{strain}} \left[2\sqrt{(f_v T_v)^2 + (f_B T_B)^2} + \sqrt{(f_v T_{2v})^2 + (f_B T_{2B})^2} \right] \exp \left[- \left(\frac{\bar{s} - \tau}{\sigma} \right)^2 \right]. \quad (\text{A.4})$$

The second term in brackets is present in the code and was absent from the earlier shortened formula. The mean score $\bar{s}$ is taken across the interface. First-jump thresholds are $\nu/\Delta x$ and $\eta/\Delta x$; the corresponding second-derivative thresholds divide those by another Δx .

The circulation coefficient combines fluid and magnetic channels, while the optional determinant channel is added separately:

$$K_p = -\sqrt{(g_{\text{rot}} f_v T_\omega)^2 + (g_{\text{mag}} f_B T_J)^2}, \quad K_{X,p} = -T_X. \quad (\text{A.5})$$

The determinant contrast uses $\sqrt{\max(0, -\det \nabla \mathbf{B})}/B_0$. There is no additional factor proportional to $|\mathbf{B}|$ in K_X : such a factor would suppress the intended response at a magnetic null. For these circulation and determinant channels, the implemented threshold keeps the absolute value if the signal reaches it anywhere; otherwise it uses a specified percentile of the signal. This conditional rule is not simply the pointwise minimum of two thresholds.

Finally, α_z is w_z times the median of nonzero magnitudes of C and K before adding K_X . It is data dependent and need not be small. The reference $w_z = 10.40$, in particular, should not be described as a small fractional perturbation. The settings used by the study are

$$\begin{aligned} \tau &= 0.1496, \quad \sigma = 0.023, \quad b_\phi = 9.94, \quad w_z = 10.40, \\ b_{\text{curl}} &= 4.27, \quad b_X = 2.39, \quad \gamma_v = 2.0, \quad \gamma_B = 0.5, \quad \kappa = 10, \quad \text{fallback percentile} = 90. \end{aligned} \quad (\text{A.6})$$

The recalibration experiment replaces τ only; the other settings remain fixed. The pipeline has nine adjustable mapping parameters when the fallback percentile is included, in addition to the separately fixed or calibrated threshold. The reference configuration is not evidence of an exhaustive search over them.

B Reproduction and provenance

The central numerical sources are listed below. Paths are relative to the pinned repository. Hyperlinks point to that revision so that the current branch cannot silently change the referenced document.

1. **Current exact and readout panels.** Tables 1 and 2 use [results/h0_recentring_summary.y.json](#) and the eight underlying `h0_optimiser_equivalence` files named in it. The circuit artifacts record clean starting revision `420404a1da5f`; the stored structural summary records `6e21bbb899c5`. The script `h0_recentring_summary.py` reads the existing circuit results and recomputes the exact structures. It does not rerun QAOA. The V1 original-threshold majority panel matches the earlier V1 artifact bit for bit in its stored energies and scores. Historical V2 solver correlations use a different normalisation and are not substituted for Table 2.

2. **Larger comparison.** Table 3 and its bootstrap intervals come from `h2b_v2_hamiltonian_vs_gbt_loso_N96_dim3_n10_multire.npz`, starting revision `3d5fe73a085a`, with a clean working tree. Its 160 recorded degeneracies are all one. The file stores fold summaries, not the full set of predictions needed to independently rerun its bootstrap. The generation script and `paired_bootstrap_delta` define how those summaries were obtained; checking stored values is distinct from generating them again.
3. **Size scan.** The reported historical numbers are read from `t26_size_scan_N256_v1.json`. Its metadata explicitly marks the starting tree as modified. It is not assigned the same status as the clean current panel.
4. **Task and numerical diagnostics.** The scripts `hard_patch_labels.py`, `dynamic_patch_labels.py` and `h1_solver_convergence.py` define the labels and convergence checks. The corresponding artifact paths and historical corrections are indexed in `docs/RESULTS.md` and qualified in `docs/EVALUATION.md`.

The additional “Exact dep.” column of Table 1 is reproduced by the accompanying `verify_results.py`. With the pinned repository as its argument and `--exact`, it reconstructs and enumerates the 48 Hamiltonians, checks their original OR scores against the stored results, and applies Eq. (2.12) directly to their binary marginals. It also confirms that all 24 V2 ground states are uniform at the spin level, not merely after the block decoder. Its machine-readable output is `verification_results.json`. This check runs no quantum circuit and changes no repository artifact.

The recentring plan can be regenerated with `study/h0_selection/h0_optimiser_equivalence.py`: four scenarios, $Re = 400$, $N = 96$, $d = 3$, three snapshots, QAOA depths 1, 2 and 3, 60 optimisation evaluations, 4096 shots and seed 0. The eight combinations use `--mapper` with `v1` or `v2`, `--bias-threshold` with `deployed` or `loso-f1`, and `--readout` with `majority` or `deployed`. The larger comparison uses `study/h2b_prediction/h2b_v2_hamiltonian_vs_gbt_loso.py`, the four Reynolds numbers and ten snapshots per scenario–Reynolds pair. Full reruns require the pinned dependencies and available DNS files. They are computational experiments, not LaTeX build steps.

Targeted regression checks include `test_h0_recentring_summary.py`, `test_h0_readout_and_bias_threshold.py` and `test_h2b_v2_hamiltonian_vs_gbt_loso_multire.py` under `tests/study/`. A passing regression check verifies its stated contract; it is not a certificate that the physical model, uncertainty analysis or complete adaptive architecture is validated.

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